

Financial Management and Valuation

Unit 2: Mini Cases

SBWL Finance

Winter Semester 2025

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **December 15, 2025 at 10pm**. To do so, go to the assignment section on Canvas and choose “Unit 2 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions file upload will be considered, your last complete attempt counts.**

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism.

The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Exercise 2.1: Valuation of a One-Period Structured Bond: The risk-free rate is 3% for the period. The Eurostoxx 50 index is currently at 5,600, and in one period it will either go up to 6,100 or go down to 5,100. A structured bond pays at maturity: EUR 1000 principal, 2% interest on the principal, plus 0.5 EUR for every 1-point increase in the Eurostoxx 50 if it goes up.

1. Draw the corresponding tree diagram.
2. Find the tracking portfolio.
3. Compute the present value of the structured bond.
4. If the bond is offered in the market at EUR 1,200 is this consistent with the law of one price?

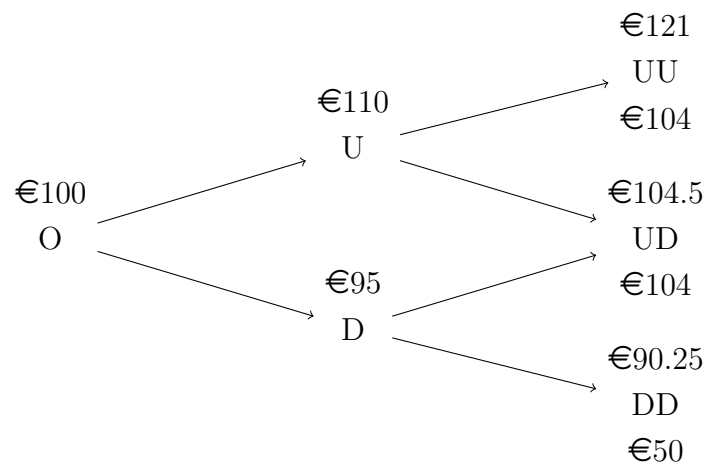
Exercise 2.2:

A structured bond has present value $V_0 = 100$. The corresponding tracking portfolio consists of 1 unit of the underlying asset and the remainder invested in a risk-free asset with unknown interest rate r_f . The current value of the underlying security is $S_0 = 120$. In the down-state, the underlying asset falls to $S_d = 95$, and the value of the tracking portfolio is observed to be $V_d = 74.2$. In the up-state, the underlying asset rises to $S_u = 140$.

First determine the value invested in the risk-free asset B . Then use the down-state information to derive the risk-free rate r_f . Finally, compute the value of the tracking portfolio in the up-state V_u .

Exercise 2.3:

Calculate the value of a risky corporate bond using the risk neutral valuation method. Assume that the risk free rate is 1% per period, with a period defined as six months. The bond has a face value of €100 payable in two periods from now.



This binomial tree provides the firm value and the cashflow of the bond below. This bond is risky in the sense that the principal may not be repaid in full in period 2, if the firm value falls below €104. What is the present value of the bond?

Exercise 2.4:

Using the simple one-period binomial approach, determine the value of a put option which has an exercise price of EUR 30, given that the stock price is EUR 28, that $u = 1.4$, that $d = 0.8$ and that $r_f = 10\%$. What is the portfolio of stocks and (risk-less zero-coupon) bonds that replicates this option?

Exercise 2.5: You are tasked with advising a client in the private banking department of a bank. The client holds an equity portfolio valued at EUR 1 million. Over the last few years, she has made substantial profits by holding tech stocks, but she is becoming increasingly concerned that the boom in AI companies has led to overvaluations. She wants to hedge against the risk of losses in these shares. You shall propose a hedging strategy based on **real** financial instruments; you will find all relevant details in the Key Information Document (KID) uploaded with the exercise. Note that the KID specifies a payout ratio of 0.01 for Option 1; it means that only one percent of the usual option payoff is actually paid, which should be reflected in your calculations.

Use the following prices for your calculations.

Product Details	Option 1	Option 2
ISIN	DE000VH5L2L0	DE000VH1GXW6
Underlying	Nasdaq-100 Index	NVIDIA Corp.
Option Type	Put	Put
Current Underlying Price	USD 25,000.00	USD 190.00
Current Option Price	EUR 6.47	EUR 13.47

1. Draw the payoff profile of the two options, with the underlying asset value on the x-axis.
2. Calculate the number of each option necessary to receive USD 10,000 in case the two underlying assets would decrease by 30%.
3. Compare the two euro amounts she would need to ensure the hedging payoff in case of a 30% decline. Which one of the two is cheaper and why?

Exercise 2.6:

Assume that the shares of the water sports company WaveUnited (WU) currently sell for 30. A European call option on WU has a strike price of 30, expires two years from now, and currently sells for 12. Assume the risk-free rate is 5% per year (annually compounded).

- (a) What is the value of a comparable European put?
- (b) Explain in your own words the intuition behind the formula in (a).
- (c) Adjust the inputs used in (a) such that the values of a European put and a European call are equal.